

Demonstration of Proposed Features

Date	03 July 2026
Team ID	N/A
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document tracks whether each feature proposed during planning was successfully implemented and can be demonstrated live.

S. No	Feature Name	Description	Status	Demonstrated	Remarks
1	Soil & Climate Input Form	Collects N, P, K, temperature, humidity, pH, rainfall from the user	Implemented	Yes	Shown live on the Find Your Crop page
2	Crop Prediction Engine	Predicts the most suitable crop using the trained Random Forest model	Implemented	Yes	Prediction shown instantly after form submission
3	Multi-Algorithm Comparison	Compares KNN, Logistic Regression, Decision Tree, and Random Forest	Implemented	Yes	Shown via notebook/train_model.py output during demo
4	Model Persistence (Pickle)	Saves and reloads the trained model/scaler without retraining	Implemented	Yes	Demonstrated by app.py loading model.pkl/scaler.pkl at startup
5	Seasonal Crop Grouping	Groups crops into Summer/Winter/Rainy based on environmental ranges	Implemented	Partial	Shown in the notebook; not yet surfaced in the web UI
6	Multi-page Navigation	Home, About, and Find Your Crop pages with shared navigation	Implemented	Yes	Demonstrated by navigating the live app

Feature Implementation Summary:

Metric	Value
Total Features Proposed	6 (from the core feature set; see also the full 8-feature list in 'No. of Functional Features')
Total Features Implemented	6
Total Features Demonstrated	5 fully, 1 partially (seasonal grouping shown in notebook only)

Metric	Value
Overall Implementation Rate (%)	100% implemented; ~92% fully demonstrated in the live app